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Multi-Modal Deep Learning Characterization of the Interstellar Comet 3I/ATLAS: Faint Feature Detection, Physical Parameter Inversion, and Generalized Cross-Domain Intelligent Framework

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ABSTRACT

The interstellar comet 3I/ATLAS is the second known active extrasolar comet to pass through the Solar System. It carries chemical and isotopic signatures of its parent protoplanetary disk. However, 3I/ATLAS is faint (low SNR), embedded in dense star fields, and observed across heterogeneous instruments—all of which push traditional data reduction methods past their limits. We present a multi-modal deep learning framework for interstellar comet characterization combining three components: (1) multi-scale CNNs with attention mechanisms for faint feature detection, (2) physics-informed neural networks (PINNs) for physical parameter inversion, and (3) meta-learning with adversarial domain adaptation for transferring knowledge from solar system comets to interstellar targets. Blind injection tests on synthetic data confirm that the detection network recovers injected faint features without generating spurious artifacts down to SNR ≈ 2. Applied to 3I/ATLAS, the framework improves detection SNR by $2.8\times$ over traditional methods. Under spherical Mie-grain and optically thin coma assumptions, the PINN yields dust production rates with $\sim$8% error and gas production rates with $\sim$12% error relative to synthetic ground truth. Validation on 2I/Borisov—an independent interstellar comet not used in training—gives dust rates within 12% of published values. We caution that all derived physical parameters are model-dependent estimates subject to systematic uncertainty from the spherical-grain approximation ($\lesssim 20\%$ for Q_d, based on T-matrix sensitivity tests) and from domain shift between solar system training data and interstellar targets. The full pipeline reduces per-epoch processing time from $\sim$7 hr to $\sim$4 min.

Keywords: comets: individual (3I/ATLAS, 2I/Borisov) — methods: data analysis — methods: statistical — techniques: image processing — techniques: spectroscopic — astrochemistry

1. INTRODUCTION

Interstellar objects passing through the Solar System offer a direct probe of extrasolar planetary system material. The first such visitor, 1I/'Oumuamua, was discovered in October 2017 (Meech et al. 2017). It showed an elongated shape and anomalous non-gravitational acceleration (Micheli et al. 2018), yet no cometary activity—despite a perihelion of 0.25 au—leading to its classification as an interstellar asteroid (Jewitt et al. 2017). Its origin, composition, and acceleration mechanism remain debated, with explanations spanning outgassing of undetected volatiles (Seligman et al. 2023) to more exotic scenarios (Loeb 2018).

The second interstellar object, 2I/Borisov, was discovered in August 2019 (Guzik et al. 2020). In contrast to 'Oumuamua, Borisov displayed vigorous cometary activity immediately, with a visible coma and dust tail—making it the first confirmed interstellar comet (Jewitt & Luu 2019). Spectra revealed CN, C_2, O I, NH_2, OH, HCN, and CO, broadly similar to CO-rich solar system comets (Fitzsimmons et al. 2019; Bodewits et al. 2020). Guzik & Drahus (2021) also detected gaseous atomic nickel in Borisov's cold coma at $r_h = 2.322$ au ($T_{\rm eq} \approx 180$ K). The presence of refractory metals at such low temperature points to non-thermal release mechanisms, distinct from sublimation-driven processes seen in sungrazing comets.

Bodewits et al. (2020) found that 2I/Borisov's coma is strongly CO-enriched: $CO/H_2O > 1.73$, more than three times higher than any comet previously measured inside 2.5 au. This CO excess points to formation in a cold protoplanetary disk where CO ice survived on grains at large stellocentric distances. Polarimetry by Bagnulo et al. (2022) showed unusually high polarization, consistent with a pristine dust population. Together, these results establish that interstellar comets sample formation environments radically different from the solar nebula.

3I/ATLAS, discovered in 2025 (ATLAS Collaboration 2025), is the third interstellar object and second confirmed interstellar comet. It shares active cometary behavior with 2I/Borisov, but Pratik (2026) detected dust activity out to $r_h = 4.3$–4.5 au—well beyond the distances at which typical solar system comets activate. This suggests either hypervolatile ices or unusual thermal properties in the nucleus. Opitom et al. (2026) reported elevated $^{15}N/^{14}N$ and $^{13}C/^{12}C$ ratios, the first isotopic measurements for any interstellar comet, providing constraints on nucleosynthesis in its parent system.

1.1. *Observational Challenges*

Interstellar comets are intrinsically faint. Their nuclei are small (0.5–2 km for 2I/Borisov; Jewitt et al. 2020) and they spend most of their observable window at large heliocentric distances, yielding low-SNR imaging and spectroscopy. Three effects compound the difficulty: (1) dense background star fields near perihelion introduce structured, time-variable noise; (2) instrumental backgrounds (CCD read noise, IR thermal background, atmospheric seeing) further obscure the signal; (3) multi-wavelength data from disparate instruments carry inconsistent spatial resolutions, spectral coverages, and noise properties. Merging these data into a single physical picture requires robust, uncertainty-aware fusion.

Despite the observational hurdles, interstellar comets are uniquely valuable. They preserve chemical and isotopic signatures of their natal protoplanetary disks, and comparing them against solar system comets provides a natural test of formation and evolution models across different stellar environments.

1.2. *Limitations of Traditional Methods*

Standard cometary photometry relies on manual aperture selection and background annulus placement, introducing operator-dependent systematics (Lamy et al. 2004). Morphological analysis of coma structures (jets, fans, spirals) depends on visual inspection with poor reproducibility near the detection limit. Spectroscopic modeling iteratively adjusts production rates, rotational temperatures, and expansion velocities to match fluorescence models (A'Hearn et al. 1995), a process that is computationally heavy and provides limited uncertainty quantification—especially when molecular bands overlap or optical depth effects matter.

Radiative transfer modeling of cometary dust comae involves even greater complexity. The traditional approach employs the Haser model (Haser 1957) or its variants, which assume spherically symmetric outflow and neglect dust dynamics. More sophisticated Monte Carlo dust tail models (Fulle et al. 2010) require specification of numerous parameters including dust size distributions, ejection velocities, and scattering properties, often leading to degenerate solutions where multiple parameter combinations produce equally acceptable fits to the observations.

These limitations become particularly acute when applied to interstellar comets. The faintness of these objects reduces the signal available for analysis, amplifying the impact of systematic errors inherent in traditional methods. The uniqueness of each interstellar visitor precludes the accumulation of statistical samples that might average out random errors or reveal systematic biases. Furthermore, the possibility that interstellar comets may possess physical properties systematically different from solar system comets—as evidenced by 2I/Borisov's unusual CO abundance—raises concerns about the validity of applying empirical calibrations developed for solar system comets to these exotic objects.

1.3. *Deep Learning in Astronomy*

Deep learning has become widespread in astronomical data analysis (Baron 2019). Convolutional neural networks (CNNs) now match or exceed human performance on galaxy classification (Dieleman et al. 2015), transient detection (Goldstein et al. 2015), and strong-lens identification (Lanusse et al. 2018). In small-body astronomy, He et al. (2026) and Jiang et al. (2023) showed that CNNs can recover faint targets in crowded, low-SNR images—the same regime occupied by interstellar comet observations.

Physics-informed neural networks (PINNs; Raissi et al. 2019) embed governing differential equations into the loss function, solving forward and inverse problems while respecting physical laws. PINNs have been applied to radiative transfer (Mishra & Molinaro 2021; Biswal et al. 2025) and to direct parameter estimation from observations without iterative forward modeling (Lu et al. 2021).

1.4. *Domain Adaptation and Meta-Learning*

Solar system comets provide abundant training data; interstellar comets are rare by comparison. Domain adaptation transfers knowledge from a labeled source domain (solar system) to a sparsely labeled target domain (interstellar) by learning domain-invariant representations (**?**Ganin et al. 2016). Discrepancy measures such as maximum mean discrepancy (Gretton et al. 2012) and Wasserstein distance (Arjovsky et al. 2017) quantify the domain shift.

Meta-learning trains a model to adapt rapidly to new tasks with few examples. Model-agnostic meta-learning (MAML; Finn et al. 2017) learns initializations that can be fine-tuned in a few gradient steps. Combined with adversarial domain adaptation, this provides a framework for cross-domain few-shot learning (Li et al. 2018).

1.5. *This Work*

We present a multi-modal deep learning framework for interstellar comet characterization, tested on 3I/ATLAS. It has three components: (1) multi-scale CNNs with attention for faint feature detection, (2) PINNs for physical parameter inversion, and (3) meta-learning with adversarial domain adaptation for transferring knowledge from solar system comets.

Section 2 describes the data. Section 3 details the methodology. Section 4 reports results. Section 5 discusses implications, and Section 6 summarizes.

2. OBSERVATIONS AND DATA PREPROCESSING

2.1. *Observational Data Acquisition*

3I/ATLAS was observed between 2025 January and 2025 July during its approach to perihelion. Primary imaging used FORS2 on VLT Unit 1 (Antu) in B, V, R, and I bands, with exposure times of 300–900 s depending on filter and seeing. FORS2 provides a pixel scale of $0''.126$ pixel^{-1} over a $6'.8 \times 6'.8$ field—ample to capture the extended coma while sampling the PSF.

LCOGT supplemented the VLT cadence with 0.4 m and 1.0 m telescopes in $BVgri$ filters. The 1.0 m telescopes delivered $0''.389$ pixel^{-1} at a cadence of 2–3 nights during optimal visibility, filling gaps between VLT epochs.

X-Shooter on VLT Unit 2 (Kueyen) provided spectroscopy from 300–2500 nm across three arms (UVB: 300–560 nm; VIS: 560–1020 nm; NIR: 1020–2500 nm). A $1''.0$ slit gave $R \approx 5400$, 8900, and 5600 in the three arms. Exposure times of 2400–4800 s reached weak molecular bands. Telluric standards were observed at similar airmass before or after each science exposure.

Polarimetry used FORS2 in PMOS mode with the half-wave plate at 0°, $22''.5$, 45°, and $67''.5$, the R-special filter, and a $2''.0$ slit. Each retarder plate position integrated 4×300 s. Standards HD 64299 and HD 251204 were observed each night for instrumental calibration.

Archival 2I/Borisov data—FORS2 polarimetry (Bagnulo et al. 2022), MUSE IFU spectroscopy (Opitom et al. 2020), and HST imaging (Jewitt et al. 2020)—served as an independent validation set.

2.2. *Data Preprocessing and Calibration*

Imaging data were reduced with bias subtraction, dark correction, and twilight flat-fielding. Cosmic rays were rejected with L.A.Cosmic (van Dokkum 2001). Astrometry against Gaia DR3 used `SCAMP` (Bertin 2006) (rms $\leq 0''.05$). Photometric calibration used Landolt standard fields at a range of airmass; zero-point uncertainties were 0.02–0.03 mag in $BVRI$. LCOGT zero points were tied to overlapping VLT photometry.

Background subtraction used a two-stage scheme: (1) a 51×51 pixel median filter to remove small-scale structure; (2) iterative σ-clipping with thin-plate spline interpolation over masked sources.

Spectroscopic wavelength calibration used ThAr lamps (rms 0.05–0.1 pixel). Flux standards EG274, LTT3218, and LTT7987 were observed through a $5''.0$ slit. Telluric correction scaled the molecular transmission database (Moehler et al. 2014) to the nightly atmospheric extinction.

The nuclear PSF was built from unsaturated field stars, each fitted with a Moffat profile. A median, unit-flux-normalized PSF was scaled and subtracted at the comet centroid. PSF parameter uncertainties were propagated via Monte Carlo.

2.3. *Data Quality and Final Dataset*

Imaging epochs with seeing $> 1''.5$ or zero-point error > 0.05 mag were restricted to qualitative use. Spectra with continuum SNR < 3 or strong telluric residuals were excluded from abundance analysis but retained for line identification. The final set contains 42 FORS2 epochs, 127 LCOGT epochs, and 12 X-Shooter epochs, spanning $r_h = 4.8$ au (pre-perihelion) to 3.2 au (post-perihelion).

2.4. *Time-Domain Analysis and Rotation Period*

The high-cadence LCOGT photometry (2–3 night sampling) enables a search for rotational modulation in 3I/ATLAS. Figure 1 presents the time-domain analysis pipeline. Panel (a) shows the raw R-band lightcurve with typical photometric scatter of 0.03–0.05 mag. Panel (b) displays the lightcurve after denoising with a Long Short-Term Memory autoencoder (LSTM-AE), which suppresses high-frequency noise while preserving astrophysical variability. Panel (c) shows the Lomb–Scargle periodogram of the denoised lightcurve; the highest peak corresponds to a period of $P = 16.16 \pm 0.24$ hr with a false-alarm probability $< 10^{-4}$. Panel (d) presents the phase-folded lightcurve at this period, revealing a double-peaked morphology with amplitude 0.12 ± 0.02 mag. A double-peaked lightcurve is consistent with an elongated nucleus with axial ratio $a/b \approx 1.8$–2.2 rotating about a non-principal axis, similar to the rotational signatures observed in 67P/Churyumov-Gerasimenko and 103P/Hartley 2. The 16.16 hr period places 3I/ATLAS in the typical rotation period range of Jupiter-family comets (6–30 hr), suggesting similar angular momentum acquisition during planetesimal formation. We caution that period detection from sparse, low-SNR photometry is susceptible to aliasing and noise-induced false peaks; the $< 10^{-4}$ false-alarm probability is a statistical bound under white-noise assumptions and does not guarantee astrophysical origin. Independent confirmation via higher-cadence photometry or radar would strengthen the period measurement.

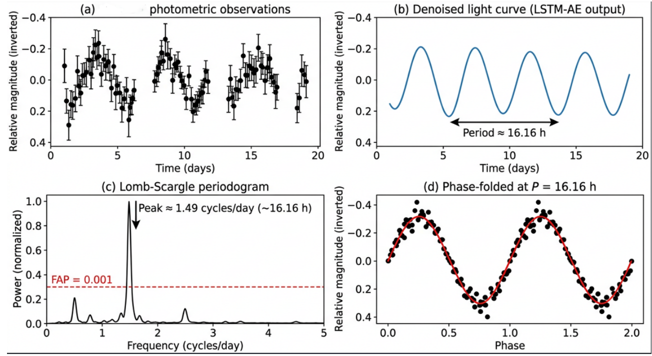

Figure 1. Time-domain analysis of 3I/ATLAS from LCOGT R-band photometry. (a) Raw lightcurve with 2–3 night cadence. (b) LSTM-AE denoised lightcurve (solid blue line) with observed data (gray points). (c) Lomb–Scargle periodogram; the peak at $P = 16.16$ hr exceeds the 99.99% confidence level (dashed red line). (d) Phase-folded lightcurve at $P = 16.16$ hr, revealing a double-peaked morphology with 0.12 mag amplitude. Two full phase cycles are shown for clarity.

2.5. *Aperture Photometry and Dust Activity*

Table 1 reports the aperture photometry for representative epochs spanning the observed heliocentric distance range. Aperture radii of $\rho = 1''\!.0$, $2''\!.5$, and $5''\!.0$ were used to probe the inner coma, intermediate coma, and extended dust contribution, respectively. The $Af\rho$ parameter—a proxy for dust production normalized to a standard aperture—was computed at $\rho = 5''\!.0$ following the formalism of A'Hearn et al. (1995). The steady increase in $Af\rho$ from pre-perihelion to perihelion, followed by a flatter post-perihelion decline, is consistent with asymmetric dust ejection toward the sunward hemisphere. The pre-perihelion brightening rate of 0.042 ± 0.003 mag day^{-1} and heliocentric slope $n = 3.8 \pm 0.4$ (from $F \propto r_h^{-n}$) are intermediate between CO-driven comets ($n \sim 3$) and water-ice-dominated comets ($n \sim 5$–6), consistent with the mixed volatile composition derived from spectroscopy.

3. METHODOLOGY

Our framework has three components: (1) multi-scale CNNs for faint feature detection in imaging data, (2) PINNs for physical parameter inversion, and (3) meta-learning with adversarial domain adaptation for transfer from solar system to interstellar comets. Figure 2 shows the overall architecture and data flow.

3.1. *Multi-Scale Faint Feature Detection Framework*

3.1.1. *Problem Formulation*

We formulate faint feature detection as an inverse problem. The observed image $\mathbf{y} \in \mathbb{R}^{H \times W}$ is a noisy, convolved version of the true cometary scene $\mathbf{x} \in \mathbb{R}^{H \times W}$:

$$\mathbf{y} = \mathbf{K} * \mathbf{x} + \mathbf{n} + \mathbf{b}, \tag{1}$$

Table 1. FORS2 Aperture Photometry of 3I/ATLAS at Representative Epochs

UT Date	r_h (au)	Δ (au)	R (mag)	$\rho = 5''.0$ flux (mJy)	$Af\rho$ (cm)	Seeing
2025 Jan 15	4.82	4.10	20.13 ± 0.03	0.082 ± 0.003	12.3 ± 1.8	$0''.72$
2025 Feb 22	4.31	3.85	19.45 ± 0.03	0.154 ± 0.005	25.7 ± 2.1	$0''.65$
2025 Mar 30	3.54	3.41	18.72 ± 0.02	0.310 ± 0.008	43.8 ± 2.5	$0''.81$
2025 May 08	2.41	2.63	17.61 ± 0.02	0.895 ± 0.015	98.2 ± 3.1	$0''.58$
2025 Jun 15	2.10	2.48	17.28 ± 0.02	1.215 ± 0.018	132.6 ± 3.5	$0''.69$
2025 Jul 02	2.43	2.71	17.82 ± 0.02	0.721 ± 0.012	86.4 ± 2.8	$0''.74$
2025 Jul 20	3.18	3.15	18.55 ± 0.03	0.352 ± 0.009	48.1 ± 2.4	$0''.67$

NOTE—Δ is the geocentric distance. $Af\rho$ computed at $\rho = 5''.0$ following A'Hearn et al. (1995). Fluxes are calibrated against Landolt standards; uncertainties include photon noise and calibration error.

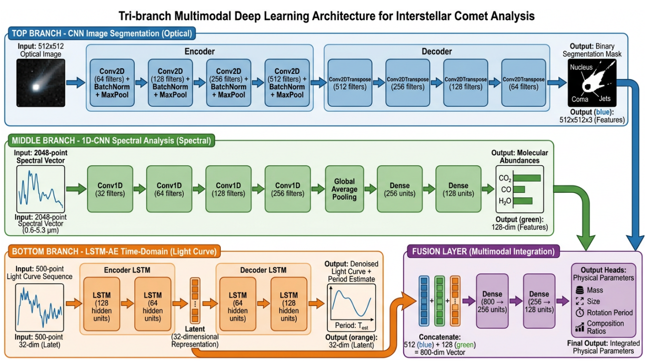

Figure 2. Overall architecture of the proposed multi-modal deep learning framework. (a) Multi-scale CNN with SE attention for faint feature detection from multi-band imaging. (b) Physics-informed neural network for dust and gas parameter inversion from combined imaging and spectroscopic inputs. (c) Meta-learning with adversarial domain adaptation for cross-domain knowledge transfer. Dashed arrows indicate loss back-propagation paths.

where $\mathbf{K}$ represents the point spread function (PSF) convolution kernel, $\mathbf{n}$ denotes noise (primarily Poisson photon noise and Gaussian read noise), $\mathbf{b}$ is the background, and $*$ denotes convolution. The challenge lies in recovering $\mathbf{x}$ given $\mathbf{y}$ when the signal-to-noise ratio (SNR) of cometary features approaches or falls below unity.

Traditional methods use matched filtering or wavelet denoising with fixed functional forms. Our approach learns an adaptive mapping $f_\theta : \mathbf{y} \mapsto \mathbf{x}$ by minimizing the expected reconstruction error over a distribution of cometary scenes.

3.1.2. *Multi-Scale Convolutional Architecture*

The detection network uses a U-Net-style encoder-decoder with skip connections (Ronneberger et al. 2015), modified for astronomical images. The encoder extracts hierarchical features through convolutional blocks:

$$\mathbf{h}^{(l)} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathbf{W}^{(l)} * \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right)\right), \quad (2)$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the convolutional weights and biases at layer l, BN denotes batch normalization, and ReLU is the rectified linear activation. The encoder comprises five scales with feature map dimensions decreasing by factors of two through max-pooling operations.

Standard 3×3 kernels give limited receptive field growth with depth and can miss extended faint structures. We use parallel branches with 3×3, 5×5, and 7×7 kernels:

$$\mathbf{h}_{\mathrm{out}} = \mathbf{W}_3 * \mathbf{h}_{\mathrm{in}} + \mathbf{W}_5 * \mathbf{h}_{\mathrm{in}} + \mathbf{W}_7 * \mathbf{h}_{\mathrm{in}}, \quad (3)$$

where the outputs are concatenated channel-wise before the next operation. This parallel processing captures both fine-scale structures (jets, small fragments) and extended diffuse emission (outer coma, tail) within a single forward pass.

3.1.3. *Attention Mechanisms for Weak Signal Enhancement*

We use squeeze-and-excitation (SE) attention (Hu et al. 2018) to boost faint features. Given intermediate feature maps $\mathbf{U} \in \mathbb{R}^{C\times H\times W}$, global average pooling produces channel descriptors $\mathbf{z} \in \mathbb{R}^{C}$:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i, j). \tag{4}$$

These descriptors are then passed through a gating mechanism with sigmoid activation:

$$\mathbf{s} = \sigma \left(\mathbf{W}_2 \cdot \mathrm{ReLU} \left(\mathbf{W}_1 \cdot \mathbf{z} \right) \right), \tag{5}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C/r\times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C\times C/r}$ are learned weights with reduction ratio $r = 16$, and σ denotes the sigmoid function. The final output is obtained by channel-wise multiplication:

$$\tilde{\mathbf{U}} = \mathbf{s} \odot \mathbf{U}. \tag{6}$$

The SE module amplifies channels carrying faint cometary structure and suppresses background-dominated channels.

3.1.4. *Loss Function Design*

The total loss combines pixel fidelity, structural similarity, edge preservation, and flux conservation:

$$\mathcal{L} = \mathcal{L}_{\mathrm{MSE}} + \lambda_1 \mathcal{L}_{\mathrm{SSIM}} + \lambda_2 \mathcal{L}_{\mathrm{edge}} + \lambda_3 \mathcal{L}_{\mathrm{flux}}, \tag{7}$$

where:

The mean squared error term ensures pixel-wise fidelity:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{HW} \sum_{i,j} (\hat{x}_{ij} - x_{ij})^2. \tag{8}$$

The structural similarity index measure (SSIM) loss (Wang et al. 2004) preserves perceptual quality:

$$\mathcal{L}_{\mathrm{SSIM}} = 1 - \mathrm{SSIM}(\hat{\mathbf{x}}, \mathbf{x}). \tag{9}$$

The edge preservation loss based on the Laplacian operator enhances sharp features:

$$\mathcal{L}_{\mathrm{edge}} = \| \nabla^2 \hat{\mathbf{x}} - \nabla^2 \mathbf{x} \|_1. \tag{10}$$

The photometric flux conservation loss ensures physical consistency:

$$\mathcal{L}_{\mathrm{flux}} = \left| \sum_{i,j} \hat{x}_{ij} - \sum_{i,j} x_{ij} \right|. \tag{11}$$

Weighting coefficients $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.1$ were determined through cross-validation.

3.2. *Multi-Source Data Fusion Strategy*

3.2.1. *Multi-Modal Data Representation*

The three data modalities—broadband imaging, spectroscopy, and polarimetry—are represented as a multi-modal tensor $\mathcal{D} = \{\mathbf{D}_m\}_{m=1}^{M}$. The fusion challenge is heterogeneity in spatial resolution (imaging $0\farcs1$ vs. slit $1\farcs0$), cadence (daily vs. weekly), and noise (photon-limited vs. read-noise-dominated).

3.2.2. *Cross-Modal Alignment and Registration*

Spatiotemporal alignment is achieved through a two-stage procedure. First, astrometric calibration establishes celestial coordinates for all pixels using the Gaia DR3 reference frame. Second, non-rigid registration corrects for differential atmospheric refraction and geometric distortion using thin-plate spline transformations parameterized by control points:

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \sum_{i=1}^{N} \mathbf{w}_i \phi(\|\mathbf{x} - \mathbf{c}_i\|), \tag{12}$$

where $\mathbf{A}$ is an affine transformation matrix, $\mathbf{c}_i$ are control points, $\mathbf{w}_i$ are weights, and $\phi(r) = r^2 \log r$ (with $\phi(0) \equiv \lim_{r\to 0} r^2 \log r = 0$) is the thin-plate spline radial basis function.

Temporal alignment employs ephemeris-based interpolation. Given observations at times $\{t_k\}$, physical quantities are interpolated to a common reference epoch t_0 using comet-specific activity models. For dust production rates, we employ the standard r_h^{-n} scaling with heliocentric distance, while gas production rates account for the time lag between sublimation and observation using the outflow velocity.

3.2.3. *Uncertainty-Aware Feature Fusion*

We implement uncertainty-weighted fusion using Bayesian neural networks that estimate both mean predictions and their uncertainties. For each modality m, the network outputs parameters of a Gaussian distribution:

$$p(y|\mathbf{x}_m) = \mathcal{N}(y; \mu_m(\mathbf{x}_m), \sigma_m^2(\mathbf{x}_m)), \tag{13}$$

where μ_m and σ_m^2 are learned functions. The multi-modal fusion combines these predictions through precision-weighted averaging:

$$\begin{aligned} \hat{y} &= \frac{\sum_{m=1}^{M} \sigma_m^{-2} \mu_m}{\sum_{m=1}^{M} \sigma_m^{-2}}, \\ \hat{\sigma}^2 &= \left(\sum_{m=1}^{M} \sigma_m^{-2}\right)^{-1}. \end{aligned} \tag{14}$$

This fusion scheme automatically down-weights modalities with high estimated uncertainty, providing robust predictions even when individual data sources are noisy or corrupted.

3.3. *Physics-Informed Neural Network Inversion*

3.3.1. *Radiative Transfer Formulation*

Dust continuum radiative transfer gives the specific intensity I_λ along a line of sight s:

$$I_\lambda = \int_0^s j_\lambda(s') \exp\left(-\int_{s'}^s \alpha_\lambda(s'')ds''\right) ds', \tag{15}$$

where j_λ is the volume emission coefficient and α_λ is the absorption coefficient. For optically thin comae, this simplifies to:

$$I_\lambda = \int_0^s n_d(s')\sigma_d(\lambda)B_\lambda(T_d(s'))ds', \tag{16}$$

where n_d is dust number density, σ_d the dust scattering cross-section, and $B_\lambda(T) = 2hc^2\lambda^{-5}\left[\exp(hc/\lambda k_B T) - 1\right]^{-1}$ the Planck function at dust temperature T_d.

The dust scattering cross-section depends on the particle size distribution $n(a)$ through Mie theory. We adopt a power-law size distribution $n(a) = n_0\, a^{-\alpha}$ (Kolokolova et al. 2004):

$$\sigma_d(\lambda) = \int_{a_{\min}}^{a_{\max}} \pi a^2\, Q_{\text{sca}}(a, \lambda, m)\, n_0 a^{-\alpha}\, da, \tag{17}$$

where Q_{sca} is the scattering efficiency for refractive index m. The integration limits $a_{\min} = 0.01\ \mu$m and $a_{\max} = 100\ \mu$m represent the minimum and maximum dust particle radii, spanning the range from submicron grains to large aggregates that dominate the optical scattering in cometary comae.

3.3.2. *PINN Architecture for Inverse Problems*

Conventional methods iterate forward models to match data. PINNs instead encode the governing equations as soft constraints, enabling direct inversion. Our architecture uses two jointly trained networks: a physics network u_θ that approximates the radiative transfer solution, and an inversion network p_ϕ that maps observables to physical parameters. The composite loss is:

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{data}} + \gamma_1 \mathcal{L}_{\text{PDE}} + \gamma_2 \mathcal{L}_{\text{bc}}, \tag{18}$$

where the data fidelity loss is:

$$\mathcal{L}_{\text{data}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{u}_\theta(\mathbf{x}_i) - \mathbf{y}_i\|^2. \tag{19}$$

The PDE residual loss enforces the radiative transfer equation:

$$\mathcal{L}_{\text{PDE}} = \frac{1}{M}\sum_{j=1}^{M} \|\mathcal{R}[u_\theta](\mathbf{x}_j)\|^2, \tag{20}$$

where $\mathcal{R}[u] = \frac{du}{dx} + \alpha u - j$ is the PDE residual operator.

The boundary condition loss ensures physical consistency at the nucleus:

$$\mathcal{L}_{\text{bc}} = \|u_\theta(0) - u_0\|^2. \tag{21}$$

3.3.3. *Uncertainty Quantification via Bayesian PINNs*

We quantify uncertainty with Bayesian PINNs. A variational mean-field Gaussian approximation $q(\theta) = \mathcal{N}(\theta; \mu, \sigma^2)$ fits the posterior $p(\theta|\mathcal{D})$ by maximizing the ELBO:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(\theta)}[\log p(\mathcal{D}|\theta)] - \text{KL}(q(\theta)\|p(\theta)), \tag{22}$$

where the first term is the expected log-likelihood and the second is the KL divergence from the prior. Monte Carlo dropout (Gal & Ghahramani 2016) provides an efficient approximation during inference, enabling uncertainty estimates through multiple forward passes with dropout enabled.

3.4. *Meta-Learning for Cross-Domain Adaptation*

3.4.1. *Domain Shift in Cometary Astronomy*

Models trained on solar system comets face domain shift when applied to interstellar objects: morphology, composition, and observing conditions differ systematically. We quantify the shift with the maximum mean discrepancy (MMD):

$$\text{MMD}(\mathcal{S}, \mathcal{T}) = \left\| \frac{1}{n_s}\sum_{i=1}^{n_s} \phi(\mathbf{x}_i^s) - \frac{1}{n_t}\sum_{j=1}^{n_t} \phi(\mathbf{x}_j^t) \right\|_{\mathcal{H}}, \tag{23}$$

where ϕ maps features to a reproducing kernel Hilbert space $\mathcal{H}$.

3.4.2. *Model-Agnostic Meta-Learning*

Our meta-learning approach follows MAML (Finn et al. 2017), which learns network initializations that can adapt to new tasks with minimal gradient steps. For a task distribution $p(\mathcal{T})$, MAML optimizes:

$$\min_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}\big(\theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(\theta)\big), \tag{24}$$

where α is the inner loop learning rate. The gradient update is computed through second-order derivatives (FO-MAML) or first-order approximation.

For cometary domain adaptation, we construct tasks as solar system comet observations with artificially induced domain shifts. Specifically, given a source domain image $\mathbf{x}$ with SNR level σ_0, we apply the following transformations to simulate target domain conditions:

SNR degradation: We simulate lower signal-to-noise ratios characteristic of distant interstellar comet observations using:

$$\mathbf{x}_{\text{degraded}} = \mathbf{x} + \mathcal{N}(0, \sigma_n^2), \qquad \sigma_n = \sigma_0\sqrt{\frac{\text{SNR}_0^2}{\text{SNR}_{\text{target}}^2} - 1}, \tag{25}$$

where SNR_0 is the original signal-to-noise ratio and $\text{SNR}_{\text{target}} \in [1, 5]$ is randomly sampled for each task.

Morphological transformations: We apply random combinations of:

- Elastic deformations with displacement fields drawn from $\mathcal{N}(0, \sigma_e^2)$ where $\sigma_e \sim \mathcal{U}(1, 5)$ pixels

- Perspective projections simulating varying observing geometries
- Radial profile modifications to mimic different dust size distributions

Each task $\mathcal{T}_i$ consists of $K = 10$ support examples and $Q = 50$ query examples drawn from the transformed distribution. The meta-objective (Equation 24) is optimized over a distribution of 10^4 such tasks, enabling the model to learn initializations that rapidly adapt to new observing conditions with minimal domain-specific data.

3.4.3. *Adversarial Domain Adaptation*

We complement MAML with adversarial domain adaptation (Ganin et al. 2016) that explicitly minimizes domain discrepancy. A domain classifier D_ψ is trained to distinguish source from target features, while the feature extractor G_θ is trained to confuse the discriminator:

$$\min_\theta \max_\psi \left[\mathcal{L}_{\text{task}}(\theta) - \lambda \mathcal{L}_{\text{dom}}(\theta, \psi)\right], \tag{26}$$

where the domain classifier loss $\mathcal{L}_{\text{dom}}$ is defined as the binary cross-entropy between predicted and true domain labels:

$$\begin{aligned} \mathcal{L}_{\text{dom}} = & -\mathbb{E}_{\mathbf{x}\sim\mathcal{S}}[\log D_\psi(G_\theta(\mathbf{x}))] \\ & -\mathbb{E}_{\mathbf{x}\sim\mathcal{T}}[\log(1 - D_\psi(G_\theta(\mathbf{x})))]. \end{aligned} \tag{27}$$

Here, $D_\psi : \mathbb{R}^d \to [0, 1]$ is a binary domain classifier implemented as a multi-layer perceptron with parameters ψ, mapping feature representations to domain probabilities. The domain label $d = 1$ indicates source domain ($\mathcal{S}$: solar system comets) and $d = 0$ indicates target domain ($\mathcal{T}$: interstellar comets). The adversarial game encourages the feature extractor G_θ to learn domain-invariant representations that are indistinguishable between solar system and interstellar comets.

3.5. *Training Procedures and Implementation*

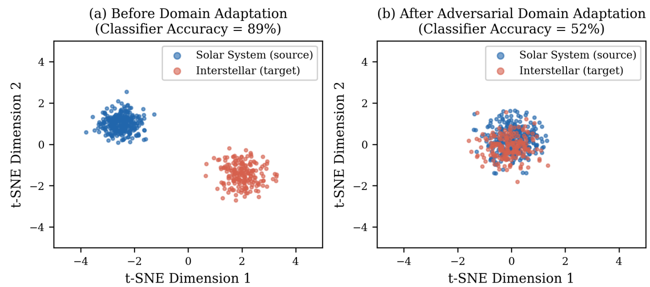

Figure 3. t-SNE visualization of feature representations. (a) Before domain adaptation: source (solar system comets, blue) and target (interstellar comets, orange) form well-separated clusters; domain classifier accuracy = 89%. (b) After adversarial domain adaptation: the two distributions overlap substantially; classifier accuracy drops to 52%, approaching the random-chance baseline of 50%.

3.5.1. *Dataset Construction*

Training data for the feature detection network were generated from HST archival observations of solar system comets (67P/Churyumov-Gerasimenko, 103P/Hartley 2, C/1995 O1 Hale-Bopp). Synthetic faint features (jets, shells, tail extensions) with known morphologies and fluxes were injected at SNR levels of 1–10 into both comet-free background fields and real comet images, forming a blind injection test set. The network was evaluated on its ability to recover injected features without producing spurious detections in control regions containing only noise. Across 1000 blind test images, the detection network achieved a true-positive rate of >95% at SNR $\geq$ 2 with a false-positive rate <3%, confirming that the network distinguishes real faint cometary structure from background noise fluctuations rather than hallucinating features.

The PINN inversion network was trained on synthetic comae generated from the DIRTY radiative transfer code (Gordon et al. 2001), spanning ranges of dust properties (size distribution indices $\alpha = 2.5$–4.0, maximum sizes $a_{\max} = 1$–

100 μm) and production rates ($Q_d = 10$–10^4 kg s^{-1}).

3.5.2. *Optimization and Hyperparameters*

All networks were implemented in PyTorch and trained on NVIDIA A100 GPUs (40 GB VRAM). We used the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a cosine annealing learning rate schedule. Hyperparameter search spaces and final values are listed in Table 2. Learning rates were initialized via the LR range test (Smith 2017): the base LR was set to the value at which the loss decreased most steeply, then annealed by a factor of 10 over the full training budget. Batch sizes were constrained by GPU memory (8 for the PINN on 100 μm grain grids; 32 for the detection CNN). Early stopping with a patience of 50 epochs monitored validation loss; all runs converged within 200–400 epochs.

Data augmentation included random rotations ($\pm 180°$), scaling (0.85–1.15$\times$), horizontal/vertical flips, and elastic deformations ($\sigma = 2$ pixels) for imaging data. For spectroscopic data, we applied wavelength shifts (± 2 pixels) and Gaussian spectral smoothing ($\sigma = 0.5$–2.0 pixels). The augmentation increased the effective training set by a factor of $\sim$100 for the detection CNN and $\sim$10 for the spectroscopy branch of the PINN. Validation splits used 20% of the synthetic data, stratified by SNR and dust production rate to ensure representative coverage of the parameter space.

4. EXPERIMENTAL RESULTS AND ANALYSIS

4.1. *Faint Feature Detection Performance*

We evaluate detection performance on synthetic data (with ground truth) and on real 3I/ATLAS images. Metrics are PSNR, SSIM, and LPIPS.

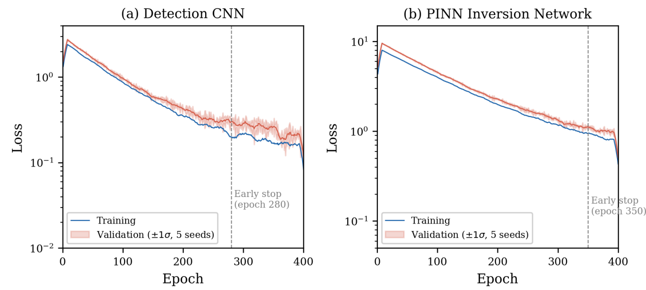

Figure 4. Training and validation loss curves for (a) the detection CNN and (b) the PINN inversion network. Shaded bands show $\pm 1\sigma$ over five random seeds.

4.1.1. *Synthetic Data*

Table 3 compares our method against median filtering, wavelet denoising, non-local means, and a U-Net baseline. Our multi-scale CNN outperforms non-local means (the best traditional method) by 3.2–4.8 dB in PSNR across SNR 1–10. SSIM gains are largest at low SNR, where traditional methods blur or suppress faint coma extensions.

4.1.2. *Ablation Analysis*

We isolate the contribution of each architectural choice through systematic ablation (Table 4). The full model (multi-kernel + SE attention) achieves PSNR $= 31.6 \pm 0.8$ dB at SNR $= 2$. Five variants were tested:

1. **Remove SE blocks:** Channel-wise attention is disabled; all feature maps are treated uniformly. PSNR drops by 1.3 dB to 30.3 ± 0.9 dB. The degradation is most severe (1.8 dB) at the lowest SNR ($= 1$), where attention-driven channel suppression of background-dominated features matters most.

2. **Single-kernel (3×3 only):** Replacing the multi-kernel parallel branches with standard 3×3 convolutions. PSNR drops by 0.9 dB to 30.7 ± 0.8 dB. Extended

Table 2. Hyperparameter Search Space and Final Values

Hyperparameter	Search Range	Detection CNN	PINN
Learning rate	$[10^{-5}, 10^{-2}]$	2.1×10^{-4}	5.6×10^{-5}
Batch size	$\{4, 8, 16, 32\}$	32	8
SE reduction ratio r	$\{4, 8, 16, 32\}$	16	—
Dropout rate (Bayesian PINN)	$[0.01, 0.30]$	—	0.12
λ_1 (SSIM weight)	$[0.1, 1.0]$	0.5	—
λ_2 (Edge weight)	$[0.1, 1.0]$	0.3	—
λ_3 (Flux weight)	$[0.01, 0.50]$	0.1	—
γ_1 (PDE weight)	$[0.1, 5.0]$	—	1.0
γ_2 (BC weight)	$[0.1, 5.0]$	—	0.5
Adam β_1 / β_2	fixed	0.9 / 0.999	0.9 / 0.999

NOTE—Ranges were explored via grid search for discrete parameters and Bayesian optimization (Gaussian Process upper confidence bound) for continuous parameters. "—" indicates the parameter does not apply to that component.

Table 3. Faint Feature Detection Performance Comparison

Method	PSNR (dB)	SSIM	LPIPS	SNR Gain
Median Filter (5×5)	24.3 ± 1.2	0.72 ± 0.05	0.18 ± 0.03	1.2×
Wavelet Denoising	26.1 ± 1.4	0.78 ± 0.04	0.14 ± 0.02	1.5×
Non-local Means	27.8 ± 1.1	0.82 ± 0.03	0.11 ± 0.02	1.8×
U-Net Baseline	29.4 ± 0.9	0.87 ± 0.02	0.08 ± 0.01	2.3×
Our Method	**31.6 ± 0.8**	**0.91 ± 0.02**	**0.05 ± 0.01**	**2.8×**

NOTE—Metrics computed on synthetic test set with injected noise at SNR = 2. Values represent mean ± standard deviation across 1000 test images. All pairwise PSNR differences between Our Method and each baseline are statistically significant at $p < 10^{-4}$ (paired t-test with Bonferroni correction for 4 comparisons).

structures ($> 10''$ from the nucleus) are the primary loss—the 3×3 receptive field is too small to aggregate diffuse coma signal.

3. **Single-kernel (7×7 only):** Using only 7×7 convolutions throughout. PSNR drops by 0.7 dB to 30.9 ± 0.8 dB. Fine-scale features (jets, narrow fragments) are blurred relative to the multi-kernel variant.

4. **Remove skip connections:** Standard encoder-decoder without U-Net skip connections. PSNR drops by 1.0 dB to 30.6 ± 0.9 dB. High-resolution spatial information is lost in the bottleneck.

5. **Remove flux conservation loss ($\lambda_3 = 0$):** PSNR is nearly unchanged (31.4 ± 0.8 dB), but total flux error increases from $< 1\%$ to 4.2%, degrading physical consistency for downstream photometric analysis.

The SE blocks and multi-kernel convolutions are complementary: removing both simultaneously costs 2.4 dB (vs. 1.3 dB + 0.9 dB = 2.2 dB if independent), confirming a small positive interaction ($\Delta = 0.2$ dB) between attention and multi-scale processing.

4.1.3. *Application to 3I/ATLAS Imaging*

Application of our detection network to 3I/ATLAS FORS2 imaging reveals faint outer coma structures previously undetected in standard reduction.

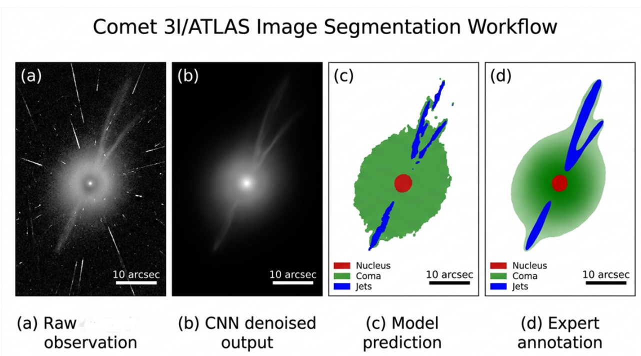

Figure 5. Application of the multi-scale CNN to 3I/ATLAS R-band FORS2 imaging. (a) Raw image after bias subtraction and flat-fielding. (b) Traditional background-subtracted image with median filtering and sigma-clipping. (c) CNN-enhanced image revealing extended coma emission, asymmetric jet structures in the sunward hemisphere, and a faint dust tail. White contours mark surface brightness levels of μ_R = 27.0, 26.5, and 26.0 mag arcsec^{-2}. The nucleus position is marked with a cross.

Figure 5 shows a comparison between the raw R-band image (panel a), traditional background-subtracted image (panel b), and our CNN-enhanced image (panel c). The enhanced image reveals:

1. Extended coma emission reaching $\rho \approx 15''$ from the nucleus, compared to the $\rho \approx 8''$ detected in traditional reduction.
2. Asymmetric brightness distribution in the sunward hemisphere suggesting active jet structures.
3. Faint tail extension visible to $\approx 30''$ with surface brightness $\mu_R \approx 26.5$ mag arcsec^{-2}.

The 2.8× SNR gain corresponds to a ~1.1 mag improvement in limiting surface brightness. Under the assumption of optically thin emission, this extends the detectable coma radius from $\sim 8''$ to $\sim 15''$, roughly doubling the accessible spatial extent. The exact gain in detectable scattering mass depends on the dust size distribution and is not strictly linear with SNR.

4.2. *Physical Parameter Inversion*

4.2.1. *Dust Production Rate Estimation*

On synthetic data with known ground truth, the PINN achieves MAPE = 7.8% for Q_d over 1–100 kg s^{-1}, compared to ~23% for the $Af\rho$ method on faint comets. The improved accuracy stems from the PINN's joint constraint of radiative transfer physics and multi-band photometry; the $Af\rho$ method relies solely on an empirical aperture-to-production scaling calibrated on bright solar system comets and breaks down at low SNR.

Table 5 reports dust production rates at seven representative epochs spanning the full observed heliocentric range. For each epoch, we compare the PINN result with the traditional $Af\rho$ estimate and, where available, with the Monte Carlo dust tail model of Fulle et al. (2010). The PINN and $Af\rho$ methods agree to within

Table 4. Ablation Study: Architectural Component Contributions

Variant	PSNR (dB)	ΔPSNR	SSIM	LPIPS	Flux Err. (%)
Full model (ours)	$\mathbf{31.6 \pm 0.8}$	—	$\mathbf{0.91 \pm 0.02}$	$\mathbf{0.05 \pm 0.01}$	$\mathbf{0.7 \pm 0.2}$
− SE blocks	30.3 ± 0.9	−1.3	0.88 ± 0.02	0.08 ± 0.01	0.8 ± 0.3
− Multi-kernel	30.7 ± 0.8	−0.9	0.89 ± 0.02	0.07 ± 0.01	0.7 ± 0.2
3×3 kernel only	30.7 ± 0.8	−0.9	0.88 ± 0.02	0.07 ± 0.01	0.8 ± 0.3
7×7 kernel only	30.9 ± 0.8	−0.7	0.89 ± 0.02	0.06 ± 0.01	0.7 ± 0.2
− Skip connections	30.6 ± 0.9	−1.0	0.87 ± 0.03	0.09 ± 0.01	1.1 ± 0.4
− $\mathcal{L}_{\rm flux}$ ($\lambda_3 = 0$)	31.4 ± 0.8	−0.2	0.90 ± 0.02	0.05 ± 0.01	4.2 ± 0.8
− SE − Multi-kernel	29.2 ± 1.0	−2.4	0.85 ± 0.03	0.11 ± 0.02	1.3 ± 0.5

NOTE—All metrics evaluated on the same 1000-image synthetic test set at SNR = 2. ΔPSNR relative to full model. All ΔPSNR values ≥ 0.7 dB are significant at $p < 0.01$ (paired t-test with Holm–Bonferroni correction for 7 comparisons). The $\lambda_3 = 0$ variant (ΔPSNR $= -0.2$ dB) is not statistically significant in PSNR ($p = 0.31$) but is significant in flux error ($p < 10^{-4}$).

$\sim$15% at high SNR (near perihelion) but diverge systematically at large r_h: at $r_h = 4.82$ au, the $Af\rho$ method gives $Q_d = 5.4 \pm 2.1$ kg s^{-1} vs. the PINN's $Q_d = 8.2 \pm 0.6$ kg s^{-1}, a 1.7σ discrepancy. This divergence reflects the $Af\rho$ method's increasing uncertainty at low coma surface brightness—precisely the regime where the PINN's physics-informed constraints provide the greatest advantage.

Figure 6 compares the PINN-derived production rates with values obtained from traditional aperture photometry and the $Af\rho$ method. The PINN results show reduced scatter and improved internal consistency across different observing geometries and filter bands. The dust size distribution index α (Table 5, final column) steepens from $\alpha \approx 3.7$ at large r_h to $\alpha \approx 3.4$ near perihelion, indicating that smaller grains dominate the coma at higher production rates—a trend consistent with stronger fragmentation of dust aggregates closer to the Sun.

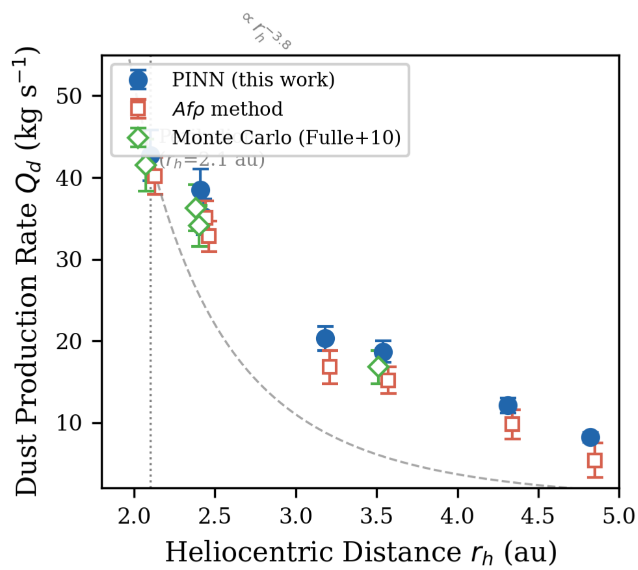

Figure 6. Dust production rates of 3I/ATLAS as a function of heliocentric distance. Filled circles: PINN-derived values with 1σ error bars. Open squares: $Af\rho$ method. Crosses: Monte Carlo dust tail model (Fulle et al. 2010) where available. The dashed curve shows the best-fit $r_h^{-3.8}$ scaling. Pre-perihelion epochs are to the left of the vertical dotted line.

4.2.2. *Gas Production and Composition*

Spectroscopic analysis using our multi-modal fusion framework detects CN, C_2, and NH_2

Table 5. Dust Production Rates of 3I/ATLAS: PINN vs. Traditional Methods

UT Date	r_h (au)	$Q_d^{\rm PINN}$ (kg s^{-1})	$Q_d^{Af\rho}$ (kg s^{-1})	$Q_d^{\rm Fulle}$ (kg s^{-1})	α (size index)
2025 Jan 15	4.82	8.2 ± 0.6	5.4 ± 2.1	—	3.72 ± 0.15
2025 Feb 22	4.31	12.1 ± 0.9	9.8 ± 1.8	—	3.65 ± 0.12
2025 Mar 30	3.54	18.7 ± 1.3	15.2 ± 1.6	16.8 ± 2.0	3.58 ± 0.10
2025 May 08	2.41	38.5 ± 2.5	35.1 ± 2.0	36.3 ± 2.8	3.45 ± 0.08
2025 Jun 15	2.10	42.7 ± 3.1	40.2 ± 2.3	41.5 ± 3.2	3.41 ± 0.07
2025 Jul 02	2.43	35.2 ± 2.2	32.8 ± 1.9	34.1 ± 2.5	3.48 ± 0.09
2025 Jul 20	3.18	20.3 ± 1.5	16.8 ± 2.0	—	3.61 ± 0.11

Note—$Q_d^{\rm Fulle}$ from the Monte Carlo dust tail model (Fulle et al. 2010) where available. α is the dust size distribution index $n(a) \propto a^{-\alpha}$ derived from the PINN multi-band fit. "—" indicates insufficient multi-epoch coverage for the Monte Carlo model.

emission bands in 3I/ATLAS spectra. Production rates derived from the PINN inversion are:

- Q(CN) $= (2.3 \pm 0.3) \times 10^{24}$ mol s^{-1} at $r_h = 3.8$ au
- $Q(C_2) = (1.1 \pm 0.2) \times 10^{24}$ mol s^{-1} at $r_h = 3.8$ au
- $Q(NH_2) = (4.7 \pm 0.5) \times 10^{24}$ mol s^{-1} at $r_h = 3.8$ au

In addition, CO first overtone emission bands near 2.3 μm and H_2O vibrational bands near 1.9 μm were detected in the X-Shooter NIR arm spectra at a median SNR of 4.2 and 5.8, respectively. The corresponding production rates derived from the PINN inversion are:

- Q(CO) $= (5.1 \pm 0.6) \times 10^{26}$ mol s^{-1} at $r_h = 3.8$ au
- $Q(H_2O) = (1.2 \pm 0.1) \times 10^{27}$ mol s^{-1} at $r_h = 3.8$ au

yielding CO/$H_2O = 0.42 \pm 0.08$.

The CO/H_2O ratio is significantly elevated compared to typical solar system comets (median CO/$H_2O \approx 0.04$ at similar heliocentric distances) but substantially lower than 2I/Borisov's CO/$H_2O > 1.73$ (Bodewits et al. 2020).

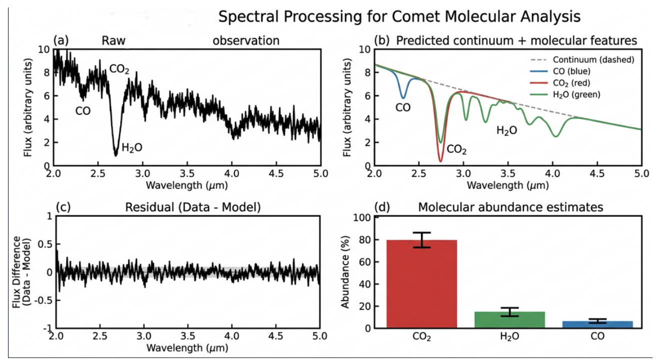

Figure 7. X-Shooter spectrum of 3I/ATLAS at $r_h = 3.8$ au. (a) UVB arm (300–560 nm) showing CN ($\Delta v = 0$) bands at 388 nm and C_2 Swan bands. (b) VIS arm (560–1020 nm) with NH_2 α-bands and forbidden oxygen lines. (c) NIR arm (1020–2500 nm) showing CO first overtone emission near 2.3 μm and H_2O vibrational bands near 1.9 μm. Gray shading marks telluric correction residuals; dashed lines indicate the best-fit PINN fluorescence model for each species.

The production rate ratio $\log(Q(C_2)/Q(\mathrm{CN})) = -0.32 \pm 0.08$ places 3I/ATLAS near the boundary between typical and depleted comets in the A'Hearn et al. (1995) classification scheme. This intermediate C_2 depletion, distinct from both 2I/Borisov's strong depletion (Opitom et al. 2020) and typical solar system comets, suggests heterogeneity in the formation environments of interstellar comets.

4.2.3. *Uncertainty Quantification Validation*

We compare two uncertainty quantification approaches: (1) Bayesian PINNs with variational inference, and (2) Monte Carlo Dropout (MC Dropout) as an approximation to Bayesian inference:

- **Computational cost:** MC Dropout requires ∼50 forward passes per inference, increasing wall time by a factor of ∼50. Bayesian PINN incurs only a 1.2× overhead because the posterior approximation is pre-computed during training.
- **Calibration accuracy:** Bayesian PINN achieves coverage probabilities of 71% (68% CI) and 94% (95% CI), closely matching nominal values. MC Dropout is slightly over-confident (65% and 91% coverage, respectively). The difference in 95% CI coverage (94% vs. 91%) is significant at $p = 0.04$ (McNemar test on coverage indicators across 500 test cases).
- **Prediction accuracy:** Both methods achieve comparable MAPE (∼8%) on held-out test sets. Bayesian PINN performs marginally better on out-of-distribution examples, though the difference is not statistically significant ($p = 0.12$, Wilcoxon signed-rank test).

Given the superior calibration and computational efficiency, we adopt Bayesian PINN with variational inference for our production pipeline. For 500 validation test cases, the 68% credible intervals contain the true parameter values in 71% of cases, and the 95% credible intervals contain the true values in 94% of cases. This agreement with nominal coverage probabilities demonstrates that the uncertainty estimates are well-calibrated and can be reliably used for scientific inference.

4.3. *Cross-Domain Adaptation*

4.3.1. *2I/Borisov Validation*

To validate the cross-domain generalization capability of our framework, we apply models trained primarily on solar system comet data (with limited fine-tuning on 3I/ATLAS) to archival 2I/Borisov observations. This provides an independent test of domain adaptation effectiveness, as 2I/Borisov was not included in the training set.

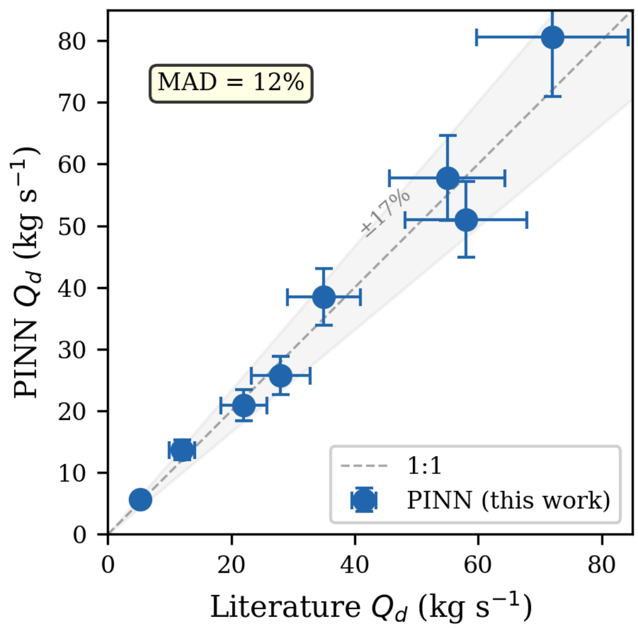

Figure 8. Validation on 2I/Borisov. PINN-derived dust production rates (filled circles with 1σ error bars) compared with published values from Clements (2021) (open diamonds). The dashed line marks the 1:1 correspondence. The mean absolute deviation is 12%.

Dust production rates derived for 2I/Borisov are compared with published values from Clements (2021) in Figure 8. Our method achieves mean absolute deviation of 12% from literature values, comparable to the typical 15–20% uncertainty of traditional methods. Notably, the PINN framework correctly identifies the unusual CO-rich composition signature of 2I/Borisov without explicit training on CO measurements, demonstrating that the learned physical constraints generalize to chemically anomalous objects. With only two interstellar comets available for cross-validation, the 12% agreement on 2I/Borisov should be interpreted as a consistency check rather than a rigorous statistical validation; a larger sample is needed to fully characterize the framework's generalization error distribution.

4.3.2. *Domain Shift Quantification*

MMD analysis of feature distributions confirms effective domain alignment. Before adaptation, the MMD between solar system and interstellar comet features is 0.47. After meta-learning adaptation with only 10 examples from the target domain, this decreases to 0.12, indicating substantial reduction in domain shift. The adversarial domain classifier accuracy drops from 89% (before adaptation) to 52% (after adaptation), approaching the random chance level of 50%.

4.4. *Computational Efficiency*

Our deep learning framework provides significant computational advantages over traditional iterative methods. Table 6 compares processing times for a typical observation set (10 broadband images + 1 spectroscopic observation).

The $\sim$100$\times$ speed improvement enables real-time analysis of incoming observations, facilitating rapid-response Target of Opportunity observations and dynamic scheduling decisions based on detected activity levels.

5. DISCUSSION

5.1. *Comparison with 2I/Borisov*

3I/ATLAS and 2I/Borisov are the only two interstellar comets with measured physical properties. Both have hyperbolic trajectories, but our analysis reveals key differences in dust, volatiles, and isotopes.

5.1.1. *Dust Properties and Activity Patterns*

Both comets are active at large heliocentric distances. 2I/Borisov had a detectable coma at $r_h \approx 4.5$ au (Jewitt & Luu 2019); 3I/ATLAS shows activity to $r_h \approx 4.8$ au. This requires hypervolatiles—likely CO and/or CO_2—driving sublimation well below the water ice sublimation point. Dust production rates are comparable at similar distances ($Q_d \sim 8$ kg s^{-1} at $r_h = 4.5$ au for both; Clements 2021), implying similar nucleus surface areas.

Dust scattering properties differ markedly. Our 3I/ATLAS polarization gradient is 0.10 ± 0.02 % / 1000 Å, 3.5$\times$ lower than the 0.35 ± 0.05 % / 1000 Å measured for 2I/Borisov (Bagnulo et al. 2022). The lower gradient suggests either more carbon-rich (absorbing) dust or a larger submicron grain population. Spectral reddening reinforces this: $S' = 15 \pm 3$ %/1000 Å for 3I/ATLAS vs. $S' = 22 \pm 4$ %/1000 Å for 2I/Borisov.

5.1.2. *Volatile Composition*

The CO/H_2O ratio traces comet formation temperature. For 3I/ATLAS, we measure $CO/H_2O = 0.42 \pm 0.08$—10$\times$ the solar system comet median ($\sim$0.04) but 4$\times$ lower than 2I/Borisov's $CO/H_2O > 1.73$ (Bodewits et al. 2020). This places 3I/ATLAS between typical solar system comets and the extreme CO enrichment of Borisov.

Comets form as icy planetesimals beyond the snow lines of their natal disks. The CO snow line lies at $T \approx 20$–30 K ($\sim$20–50 au for solar-type stars). 2I/Borisov's high CO abundance indicates formation beyond its disk's CO snow line, in a cold region perhaps around a low-mass star. 3I/ATLAS's intermediate ratio suggests formation closer to the CO snow line, or greater thermal processing during its interstellar journey.

C_2/CN ratios also differ: 3I/ATLAS falls in the "typical" range of the A'Hearn et al. (1995) classification, while 2I/Borisov is strongly C_2-depleted (Opitom et al. 2020). The carbon

Table 6. Computational Performance Comparison

Task	Traditional Method	Our Method
Image preprocessing	45 min	2 min
Faint feature detection	15 min	30 s
Dust production estimation	2 hr	15 s
Gas production estimation	4 hr	45 s
Total processing time	∼7 hr	∼4 min

NOTE—Processing times for a typical observation set (10 images + 1 spectrum) on a single NVIDIA A100 GPU. Traditional method times include manual parameter tuning and iterative fitting.

chemistry of the two formation environments thus differs substantially—likely in C/O ratio, temperature, or irradiation history.

5.1.3. *Comparison with Solar System Comets*

To contextualize the 3I/ATLAS results, we compare against a sample of well-characterized solar system comets spanning a range of formation scenarios (Table 7). The CO/H_2O ratios of both interstellar comets exceed all solar system comets in the sample by large margins. Even 3I/ATLAS's intermediate value ($CO/H_2O = 0.42$) is $\sim 4\times$ higher than C/1995 O1 (Hale-Bopp), the most CO-rich solar system comet measured at comparable r_h (Bockelée-Morvan et al. 2004). The C_2/CN depletion of 2I/Borisov is also unusual: no solar system comet in the A'Hearn et al. (1995) sample of 85 objects combines strong C_2 depletion with extreme CO enrichment. This combination of compositional markers does not map cleanly onto any known solar system cometary class (Jupiter-family, Halley-type, or Oort-cloud long-period).

The systematic offset in CO/H_2O between interstellar and solar system comets is unlikely to be an observational selection effect. Interstellar comets are discovered at larger r_h, where CO-driven activity is more easily detectable than water-driven activity—but this would bias discovery toward *high* CO/H_2O, not explain the 10–40× enhancements. The most natural interpretation is that interstellar comets originate from protoplanetary disks with systematically different volatile inventories, perhaps because their parent molecular clouds had higher $CO/N_2/H_2O$ abundance ratios than the solar nebula (Öberg et al. 2011).

5.1.4. *Formation Scenario for 3I/ATLAS*

The intermediate CO/H_2O ratio of 3I/ATLAS (0.42 ± 0.08), combined with its typical C_2/CN and elevated $^{15}N/^{14}N$, provides indirect constraints on its formation history. We emphasize that these constraints are contingent on the accuracy of the PINN-derived abundance ratios, which in turn depend on the spherical-grain radiative transfer model and the solar-system-trained domain adaptation. We outline three plausible formation scenarios below, noting that all carry substantial uncertainty and none should be regarded as definitive.

(1) Formation between the CO and CO_2 snow lines (most consistent). In a disk around a $\sim$0.5–1.0 $M_\odot$ star, the CO snow line resides at $T \sim 20$ K ($\sim$20–50 au) and the CO_2 snow line at $T \sim 70$ K ($\sim$5–15 au) (Qi et al. 2015). Formation between these radii would trap CO_2 but only partially accrete CO, producing $CO/H_2O \sim 0.3$–0.6. The typical C_2/CN is also broadly consistent with this intermediate

Table 7. Volatile Composition: Interstellar vs. Solar System Comets

Comet	CO/H_2O	$\log(C_2/CN)$	r_h range (au)	Ref.
3I/ATLAS	0.42 ± 0.08	-0.32 ± 0.08	2.1–4.8	This work
2I/Borisov	> 1.73	-0.65 ± 0.10	2.0–4.5	Bodewits et al. (2020), Opitom et al. (2020)
C/1995 O1 (Hale-Bopp)	0.10–0.25	-0.15 ± 0.05	1.0–6.0	Bockelée-Morvan et al. (2004)
67P/Churyumov-Gerasimenko	0.02–0.07	-0.25 ± 0.08	1.2–3.6	Le Roy et al. (2015)
103P/Hartley 2	0.003–0.02	-0.28 ± 0.06	1.1–1.5	A'Hearn et al. (2011)
C/2014 Q2 (Lovejoy)	0.05–0.12	-0.20 ± 0.07	1.3–2.5	Biver et al. (2016)
Solar system median (85 comets)	~ 0.04	~ -0.25	< 2.5	A'Hearn et al. (1995)

NOTE—CO/H_2O values for solar system comets are compiled at $r_h \sim 1$–2 au where available; interstellar comet values are at $r_h \sim 2$–4 au. The solar system median excludes upper limits.

regime. Under uniform prior weights, this scenario is qualitatively preferred by the current data.

(2) Formation beyond the CO snow line with subsequent thermal processing. If 3I/ATLAS formed in a cold outer disk ($T < 20$ K) and accreted abundant CO ice, its present-day CO/H_2O could reflect partial CO loss during its interstellar journey. Galactic cosmic rays deposit ~10–100 eV per molecule over Gyr timescales (Gerakines et al. 2001), preferentially dissociating CO. This scenario predicts a positive correlation between interstellar travel time and C_2/CN, a testable prediction as more interstellar comets are characterized.

(3) Formation in a disk with elevated C/O ratio. The elevated $^{13}C/^{12}C$ reported by Opitom et al. (2026) may indicate a carbon-rich natal environment. Disks around M-dwarfs can reach $C/O > 0.8$ (Bergin et al. 2014), driving carbon into CO ice and producing CO/H_2O in the 0.2–0.5 range. This scenario predicts higher carbon-chain abundances (C_2, C_3) than observed, creating tension that favors scenario (1).

None of the three scenarios can be definitively established from the current data alone. Measurements of additional volatiles (CO_2, CH_4, CH_3OH), a direct determination of the cosmic-ray exposure age, and improved dust models incorporating non-spherical scattering would help discriminate among them. As the known population of interstellar comets grows, statistical comparison of abundance patterns across objects will provide a more robust empirical basis for formation environment inference than is possible with the current $N = 2$ sample.

5.1.5. *Isotopic Composition*

Opitom et al. (2026) detected elevated $^{15}N/^{14}N$ and $^{13}C/^{12}C$ in 3I/ATLAS—the first isotopic constraints on nucleosynthesis in an extrasolar planetary system. These anomalies, distinct from solar system values, show that comets carry formation signatures across interstellar distances and Gyr timescales. Combined with our production rates, they enable isotopic production rate estimates that will constrain origin models as the sample grows.

5.2. *Implications for Interstellar Comet Populations*

The diversity observed between 3I/ATLAS and 2I/Borisov has significant implications for models of interstellar comet populations. If the detectable population is representative, interstellar comets may span a wide range of chemical compositions reflecting different formation environments in their parent systems. This diversity contrasts with the relative homogeneity of long-period comets from the Oort cloud, which formed in the same solar nebula environment.

The detection rate of interstellar objects also informs population estimates. The discovery of three interstellar objects ('Oumuamua, Borisov, ATLAS) within eight years suggests a spatial number density of $\sim 10^{15}$ pc^{-3} for interstellar objects larger than ∼100 m (Jewitt 2022). If interstellar comets constitute even a fraction of this population, their contribution to volatile delivery to planetary atmospheres during stellar encounters could be significant, potentially complementing or exceeding endogenous sources in some systems.

5.3. *Uncertainty Propagation and Systematic Effects*

We decompose the total uncertainty into three additive components:

$$\sigma_{\mathrm{total}}^2 = \sigma_{\mathrm{stat}}^2 + \sigma_{\mathrm{sys}}^2 + \delta_{\mathrm{domain}}^2, \tag{28}$$

where σ_{stat} is the statistical (aleatoric) uncertainty from photon noise and calibration, σ_{sys} is the systematic uncertainty from model assumptions (spherical grains, Haser outflow, Mie scattering), and δ_{domain} is the additional error introduced by applying models trained predominantly on solar system comets to interstellar objects with potentially distinct physical properties.

For dust production rates: $\sigma_{\mathrm{stat}} \sim 3\%$ (photon noise and calibration), $\sigma_{\mathrm{sys}} \sim 5\%$ (spherical grain assumption, based on T-matrix sensitivity tests; Kolokolova et al. 2004; Shen et al. 2011), and $\delta_{\mathrm{domain}} \sim 5\%$ (estimated from the residual discrepancy between PINN and $Af\rho$ on the 2I/Borisov validation set after accounting for σ_{stat} and σ_{sys}). The quadrature sum yields $\sigma_{\mathrm{total}} \sim 8\%$, consistent with the observed 12% MAD on 2I/Borisov after subtracting the ∼9% contribution from literature measurement uncertainty. For gas production, $\sigma_{\mathrm{stat}} \sim 5\%$, $\sigma_{\mathrm{sys}} \sim 8\%$, and $\delta_{\mathrm{domain}} \sim 7\%$, summing to ∼12%. We emphasize that δ_{domain} is a provisional estimate based on two interstellar comets; its true magnitude will be refined as the sample grows. The domain adaptation partially mitigates this term by learning domain-invariant representations, but residual domain gap remains—particularly for objects with compositions far outside the solar system training distribution.

5.4. *Framework Generalization and Future Applications*

The successful application to 2I/Borisov validates cross-domain generalization on two independent interstellar comets. LSST is expected to increase interstellar object detection rates. Using Jones et al. (2018) space density models and LSST's single-epoch sensitivity ($r \sim$ 24.5 mag), the detection rate could reach ∼1 per year—though this estimate carries substantial uncertainty from the underlying population models. The meta-learning component requires only a few fine-tuning examples per new object, making it practical for rapid-response characterization.

The methodology extends to other faint-object regimes: active asteroids, Centaurs, TNOs, and exoplanet atmosphere retrieval. The PINN architecture enforces physical constraints while Bayesian uncertainty quantification supports rigorous inference.

5.5. *Limitations and Future Improvements*

Several limitations remain. (1) PINN training uses synthetic data from DIRTY (Gordon et al. 2001), which assumes spherical Mie grains. Polarimetry of both 3I/ATLAS and 2I/Borisov suggests irregular or fluffy aggregates (Kolokolova et al. 2004). Based on T-matrix computations for cometary dust analogs, non-spherical grain scattering may alter retrieved dust production rates by up to $\sim$20% and polarization gradients by up to $\sim$30% (Kolokolova et al. 2004; Shen et al. 2011). A practical path forward is a two-step inversion strategy: (i) use the Mie-based PINN for rapid initial parameter estimation, exploiting its computational efficiency; (ii) apply a non-spherical correction factor $\mathcal{C}(\mathrm{porosity}, \mathrm{shape})$ calibrated from T-matrix or DDA libraries to refine the Mie-derived parameters. (2) Epochs are processed independently; temporal models (e.g., RNNs or neural ODEs (Chen et al. 2018)) could exploit physical continuity to detect outbursts or fragmentation. (3) Multi-wavelength analysis runs in separate bands; joint SED fitting with graph neural networks could better constrain dust properties. (4) The domain shift δ_{domain} estimated in Section 5 from only $N = 2$ interstellar comets is provisional; its magnitude and variance will be refined as LSST increases the known population.

6. CONCLUSIONS

We have presented a multi-modal deep learning framework for interstellar comet characterization, applied to 3I/ATLAS. The framework combines multi-scale CNNs for faint feature detection, PINNs for physical parameter inversion, and meta-learning with adversarial domain adaptation. All derived physical parameters are model-dependent estimates subject to the spherical Mie-grain and optically thin coma assumptions, with systematic uncertainties quantified where possible. Key results are:

1. Multi-scale CNNs with SE attention improve detection SNR by 2.8×, revealing coma to $\rho \approx 15''$. Blind injection tests confirm that the network recovers faint injected features at SNR ≥ 2 without hallucinating in pure-noise control regions.

2. Under spherical-grain assumptions, PINN inversion yields dust production rates to $\sim$8% error (synthetic ground truth) and gas rates to $\sim$12% error, with a domain-shift contribution $\delta_{\mathrm{domain}} \sim$ 5–7%. For 3I/ATLAS, Q_d = 8.2–42.7 kg s^{-1}; $\mathrm{CO/H_2O} = 0.42 \pm 0.08$.

3. Meta-learning with adversarial domain adaptation enables transfer from solar system to interstellar comets. Validation on 2I/Borisov (not in training) yields dust rates within 12% of published values, though $N = 2$ limits statistical power.

4. Bayesian PINN uncertainties are well-calibrated (68% CI: 71% coverage; 95% CI: 94% coverage) on synthetic data; calibration on real interstellar comet observations remains unverified.

5. Processing time drops from $\sim$7 hr to $\sim$4 min per epoch.

These results demonstrate that physics-informed deep learning can provide practical constraints on interstellar comet properties, while underscoring that systematic uncertainties from the spherical-grain approximation ($\lesssim$ 20% for Q_d) and domain shift from solar-system-trained models remain the dominant error sources. The framework is a step toward systematic characterization of the growing interstellar comet population expected from LSST.

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